Valid types of switch expression:
----
byte
short
int
char
Byte
Short
Integer
Character
enum (JDK 1.5)
String (JDK 1.7)

Invalid types are:
----
long
float
double
boolean

Some important points:
----
a) We cannot accept **duplicate** case value.

b) The label of case must be **constant OR literals OR expression**.

c) The break statement is used to exit from the switch block. It is optional but recommended [JDK 17 (->) break is not required]

d) The default case is **optional** and executes if no case matches in the switch expression as well as we can write default case anywhere in the switch.

//Programs on switch case :

SwitchDemo.java [Old Style]

```
 //switch case

 void main()
 {
      int choice = Integer.parseInt(IO.readln("Enter your choice :"));

      switch(choice)
      {
           case 1 :
                IO.println("Case 1");
           break;
           case 2 :
                IO.println("Case 2");
           break;
           case 3 :
                IO.println("Case 3");
           break;
           default : IO.println("Invalid Case");
      }
 }
```
SwitchDemo1.java

```
/* Write a switch case program where based on the first character
select the color  */

void main()
{
   char ch = IO.readln("Enter Color Name :").toUpperCase().charAt(0);

   switch(ch)
     {
         case 'R' -> IO.println("RED Color");
         case 'W' -> IO.println("WHITE Color");
         case 'B' -> IO.println("BLUE Color");
         case 'G' -> IO.println("GREEN Color");
         default -> IO.println("Invalid color choice");
     }

     IO.println("Out of switch body");
}
```
SwitchDemo2.java

```
 void main()
  {

        IO.println("\t\t**Main Menu**\n");
          IO.println("\t\t**100 Police**\n");
          IO.println("\t\t**101 Fire**\n");
          IO.println("\t\t**102 Ambulance**\n");
          IO.println("\t\t**139 Railway**\n");
          IO.println("\t\t**181 Women's Helpline**\n");

          IO.print("Enter your choice :");
          int choice = Integer.parseInt(IO.readln());

          switch(choice)
          {
          case 100 ->  IO.println("Police Services");

          case 101 ->  IO.println("Fire Services");

          case 102 -> IO.println("Ambulance Services");

          case 139 -> IO.println("Railway Enquiry");

          case 181 -> IO.println("Women's Helpline ");

          default ->     IO.println("Your choice is wrong");

      }
  }
```
The same program we can re-write with the help of an infinte loop to display the menu.
```
 void main()
  {
    while(true)
      {
           IO.println("\t\t**Main Menu**\n");
           IO.println("\t\t**100 Police**\n");
           IO.println("\t\t**101 Fire**\n");
           IO.println("\t\t**102 Ambulance**\n");
           IO.println("\t\t**139 Railway**\n");
           IO.println("\t\t**181 Women's Helpline**\n");

           IO.print("Enter your choice :");
           int choice = Integer.parseInt(IO.readln());

           switch(choice)
           {
           case 100 ->  IO.println("Police Services");

           case 101 ->  IO.println("Fire Services");

           case 102 -> IO.println("Ambulance Services");

           case 139 -> IO.println("Railway Enquiry");

           case 181 -> IO.println("Women's Helpline ");

           default ->     IO.println("Your choice is wrong");

      }

     String userChoice = IO.readln("Do u want to continue [yes/no] :");
       if(userChoice.equalsIgnoreCase("no"))
            break;


      }

  }
```
SwitchDemo3.java

```
 //Passing String in the switch case [Allowed from JDK 1.7]

 void main()
 {
      String season = IO.readln("Enter the name of the season :").toLowerCase();

      switch(season)  //JDK 1.7
      {
           case "summer" -> IO.println("Summer season");
           case "rainy" -> IO.println("Rainy season");
           case "winter" -> IO.println("Winter season");
           case "spring" -> IO.println("spring season");
           default -> IO.println("Invalid season");
      }

 }
```
Switch Case with Grouping in Java:

In switch statement, It is possible to combine **multiple case labels** to execute one common block of code. It must be separated by , (comma)

We should use when **multiple values (cases)** should result in the same action.

Grouping.java

```
 //switch with grouping

 void main()
 {
      int choice = Integer.parseInt(IO.readln("Enter your choice 1 - 10 :"));

      switch(choice)
      {
           case 1,3,5,7,9 -> IO.println("Odd Number");
           case 2,4,6,8,10 -> IO.println("Even Number");
      }
 }
```
SwitchDemo4.java

```
//Grouping
void main()
{
     IO.print("Enter an animal Name :");
     String animal = IO.readln().toLowerCase();

   switch(animal)
     {
          case "cat", "dog", "cow" ->
          {
               IO.println("Domestic Animal");
          }

          case "lion", "tiger", "fox" ->
          {
         IO.println("Wild Animal");
          }

     }

}
```
Switch Case with Return Value (Expression Style):

From JDK 14V onwards, A switch statement can also **return some value** that means we can use switch statement as **switch expression**.

This is known as switch case with return value OR expression style switch.

Rules:
----
a) While writing switch body ; (semicolon is compulsory at the end)
b) Here default -> is compulsory [If no case will match then default will return value]
c) From the switch body, we need to return **appropriate value based on type**.
d) If, we have multiple statements ({} using block) then
   yield keyword is compulsory to return the value explicitly
   [In a single statement, value will return automatically]

SwitchDemo5.java

```
//Based on the student grade return the student result

void main()
{
  char grade = IO.readln("Enter the grade :").toUpperCase().charAt(0);

   String result = switch(grade)
     {
     case 'A', 'B' -> "Excellent!!!";
       case 'C' -> "Very Good!!!";
     case 'D' -> "Good!!!";
     default -> "Invalid Grade";
     };

     IO.println("Your grade is :"+grade+" and your result is :"+result);

}
```
Case 1 :

```
//Based on the student grade return the student result

void main()
{
  char grade = IO.readln("Enter the grade :").toUpperCase().charAt(0);

   String result = switch(grade)
     {
     case 'A', 'B' -> "Excellent!!!";
       case 'C' -> "Very Good!!!";
     case 'D' -> "Good!!!";
     default -> "Invalid Grade";
     }  //Semicolon is compulsory here

     IO.println("Your grade is :"+grade+" and your result is :"+result);

}
```
Case 2 :

```
void main()
{
  char grade = IO.readln("Enter the grade :").toUpperCase().charAt(0);

   String result = switch(grade)
     {
     case 'A', 'B' -> "Excellent!!!";
       case 'C' -> "Very Good!!!";
     case 'D' -> "Good!!!";
     default -> "Invalid Grade";  //Writing defult is compulsory
     };

     IO.println("Your grade is :"+grade+" and your result is :"+result);

}
```
Case 3 :

```
void main()
{
  char grade = IO.readln("Enter the grade :").toUpperCase().charAt(0);

   String result = switch(grade)        Automatically will assign to result which is of String type
     {                                  for a SINGLE STATEMENT
     case 'A', 'B' -> "Excellent!!!";
       case 'C' -> "Very Good!!!";
     case 'D' -> "Good!!!";
     default -> "Invalid Grade";
     };

     IO.println("Your grade is :"+grade+" and your result is :"+result);

}
```
Case 4 :

```
 //Based on the student grade return the student result

 void main()
 {
   char grade = IO.readln("Enter the grade :").toUpperCase().charAt(0);

    String result = switch(grade)
      {
      case 'A', 'B' ->
           {
            yield "Excellent!!!";  //{} reprsents multiple statements so yield keyword is compulsory to return the
                                                                       value explicitly
           }
        case 'C' -> "Very Good!!!";
      case 'D' -> "Good!!!";
      default -> "Invalid Grade";
      };

      IO.println("Your grade is :"+grade+" and your result is :"+result);

 }
```
Case 5:

```
 //Based on the student grade return the student result

 void main()
 {
   char grade = IO.readln("Enter the grade :").toUpperCase().charAt(0);

    String result = switch(grade)
      {
      case 'A', 'B' -> yield "Excellent!!!";  //for single statement yield is not reqd
        case 'C' -> "Very Good!!!";
      case 'D' -> "Good!!!";
      default -> "Invalid Grade";
      };

      IO.println("Your grade is :"+grade+" and your result is :"+result);

 }
```
Case 5

```
 //Based on the student grade return the student result

 void main()
 {
   char grade = IO.readln("Enter the grade :").toUpperCase().charAt(0);

    String result = switch(grade)
      {
      case 'A', 'B' ->
           {
             yield "Excellent!!!";
             yield "Super!!!";  //Unreachable only one yield is allowed
           }
        case 'C' -> "Very Good!!!";
      case 'D' -> "Good!!!";
      default -> "Invalid Grade";
      };

      IO.println("Your grade is :"+grade+" and your result is :"+result);

 }
```